Optomechanical LiFi Sensor

Complete System Build Reference

All software in C++. WGM microsphere included with full lab procedure. MHz optical detection PCB included.

System Capability Summary

Digitiser tiers: (1) MCP3208 at 50 kSPS via pigpio DMA — 25 kHz Nyquist, all optical intensity and CT current modules. (2) Red Pitaya STEMlab 125-14 at 125 MSPS — 62.5 MHz Nyquist, Module 7 MHz optical detection PCB. (3) RTL-SDR — independent RF path, 100 kHz–1.766 GHz.

Module	Measures	Bandwidth	Status
Ch. A — Mode A	Visible intensity, slow	DC – 16 Hz lowpass	Functional
Ch. A — Mode B	Bandpass optical, audio	Tunable 160 Hz – 16 kHz	Functional
Ch. A — Mode C	Highpass, wide analog BW	Analog Fc ≈1.6 MHz; MCP3208 captures DC–25 kHz	Functional (Module 7 for >25 kHz)
Ch. B passive	Near-IR intensity	DC – 10 kHz	Functional
FP cavity	Acoustic fringe modulation	DC – 10 kHz acoustic envelope	Functional
WGM microsphere	Resonance-enhanced acoustic / perturbation	Sub-Hz to acoustic band	Functional (lab equipment required)
Module 7 MHz PCB	Optical 1–100 MHz	DC – 62.5 MHz (Red Pitaya 125 MSPS)	Functional
RTL-SDR	RF spectrum	100 kHz – 1.766 GHz, 2.4 MHz IF BW	Functional
Power line CT	Mains current, harmonics	DC – 10 kHz	Functional
LiFi OOK	Optical data link	4800–9600 baud decoded	Functional

HARDWARE LIMIT: Python SPI polling: 5–8 kSPS. All C++ code in this document uses pigpio DMA SPI at 40–50 kSPS. Compile: g++ -O2 file.cpp -lpigpio -lrt -o binary. Run as root or with sudo. Module 7 uses Red Pitaya C++ SCPI API; compile on any Linux host with access to the Red Pitaya network address.

Shared Hardware — Modules 1, 2, 3, 5

One RPi Zero 2W, one MCP3208, one 16 GB microSD, shared across all analog modules.

1. Raspberry Pi Zero 2W Setup

1. Flash Raspberry Pi OS Lite 64-bit. Enable SSH and SPI in imager pre-config.
2. Boot and install:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y pigpio pigpio-tools libpigpio-dev g++ cmake libi2c-dev
sudo raspi-config  # Interface Options → SPI → Enable, I2C → Enable
sudo systemctl enable pigpiod && sudo systemctl start pigpiod
pigs t   # verify: returns tick count
```

2. MCP3208 Wiring

Parameter	Value	Notes
Pin 1 (CS)	RPi GPIO8 (CE0)	
Pin 2 (Din)	RPi GPIO10 (MOSI)	
Pin 3 (Dout)	RPi GPIO9 (MISO)	
Pin 4 (CLK)	RPi GPIO11 (SCLK)	
Pin 9 (DGND)	GND	
Pin 15 (Vref)	+3.3 V	100 nF ceramic to GND at pin
Pin 16 (Vdd)	+3.3 V	100 nF ceramic to GND at pin

Note: Max SPI clock at 3.3 V: 1.5 MHz. Set spiOpen speed to 1000000.

3. Power Supply

- +5 V analog: 7805 TO-220 from 9 V wall supply. 470 µF + 100 nF at output.
- Mid-rail +2.5 V: two 10 kΩ, +5 V to GND, midpoint buffered by OPA2354 Ch.B unity-gain follower. 10 µF + 100 nF to GND at output.
- RPi: separate 5 V USB supply. Do not share with analog rail.
- Verify mid-rail: 2.500 V ±0.050 V before powering any signal board.

Component	Qty	Source	USD ea.	Notes

Raspberry Pi Zero 2W	1	raspberrypi.com / Adafruit	$15.00	
MCP3208-CI/P DIP	1	Digi-Key	$3.50	
16 GB microSD Class 10	1	Any	$8.00	
7805 TO-220	1	Digi-Key	$0.60	
470 µF / 25 V electrolytic	2	Digi-Key	$0.35 ea.	
100 nF ceramic (50 pcs)	1 bag	AliExpress	$3.00	
10 kΩ 1% (100 pcs)	1 bag	AliExpress	$2.00	
9 V 1 A wall supply	1	Amazon	$8.00	Centre-positive
Perfboard 10×15 cm	3	AliExpress	$2.50 ea.	
22 AWG hookup wire assorted	1	Amazon	$7.00	
Subtotal			~$55	

Module 1 — Channel A: Visible Spectrum Detector

Visible light intensity and modulation, DC to ~10 kHz, three filter modes via rotary switch, output MCP3208 CH0.

1.1 Parts

Component	Qty	Source	USD ea.	Notes
OPA2354 DIP8	1	Digi-Key	$2.50	Ch.A = TIA, Ch.B = mid-rail buffer
LM358 DIP8	1	Digi-Key	$0.50	Sallen-Key filter op-amp
BPW34 silicon photodiode	1	Digi-Key	$1.20	35 mm², 550 nm peak
25 mm biconvex lens f=50 mm	1	AliExpress	$4.00	
30 mm OD lens tube ~60 mm	1	AliExpress	$5.00	Spray interior matte black
3-pole 3-position rotary switch	1	Digi-Key	$3.20	
1 MΩ resistor 1%	1	Digi-Key	$0.10	Rf Mode A
100 kΩ resistor 1%	2	Digi-Key	$0.10	Rf Mode B; SK R2
1 kΩ resistor 1%	4	Digi-Key	$0.05	
10 kΩ resistor 1%	4	Digi-Key	$0.05	

100 kΩ pot linear	1	Digi-Key	$1.80	SK Mode B R1 tune
2.2 pF COG ceramic	1	Digi-Key	$0.15	Cf Mode A
1 pF COG ceramic	1	Digi-Key	$0.20	Cf Mode B
100 pF ceramic	1	Digi-Key	$0.08	Mode C Cf
10 nF film cap	2	Digi-Key	$0.20	SK C1
100 nF film cap	4	Digi-Key	$0.25	SK C2 + decoupling
1 µF film cap	1	Digi-Key	$0.40	Mode A lowpass C
Enclosure 100×60×40 mm	1	AliExpress	$6.00	
Subtotal			~$32	

1.2 TIA Stage

Stability — minimum Cf

Cf >= sqrt(Cin / (2*pi * GBW * Rf))

Mode A Rf=1MΩ: Cin=23pF, GBW=100MHz → Cf_min = 1.9 pF → use 2.2 pF

Mode B Rf=100k: Cf_min = 0.6 pF → use 1 pF

Mode C Rf=1k: no Cf needed (parasitic sufficient)

Wiring

BPW34 cathode (short lead) → +5 V

BPW34 anode → OPA2354 Pin 2 (inv.) [trace <5 mm, shielded]

OPA2354 Pin 3 (non-inv.) → mid-rail 2.5 V

OPA2354 Pin 4 (V−) → GND

OPA2354 Pin 8 (V+) → +5 V [100 nF to GND at pin]

Rf + Cf in parallel: Pin 1 back to Pin 2

OPA2354 Pin 6 (inv. B) → Pin 5 (out B) [unity-gain mid-rail buffer]

OPA2354 Pin 7 (non-inv. B) → mid-rail divider

1.3 Filter Modes

Mode A — Lowpass (Rf=1 MΩ, Cf=2.2 pF, R=10 kΩ, C=1 µF, Fc≈16 Hz)

Mode B — Sallen-Key Bandpass

Dedicated LM358 stage. TIA output feeds SK input; do not share op-amps.

TIA output → R1 (100k pot) → node X → R2 (1k) → LM358 non-inv. (+)

node X → C1 (10 nF) → LM358 output

C2 (100 nF): LM358 non-inv. → GND

LM358 inv. → LM358 output [unity-gain]

f0 = 1/(2*pi*sqrt(R1*R2*C1*C2)) range 160 Hz – 16 kHz

Q = 0.5 * sqrt(R2*C1 / (R1*C2)) range 1.6 – 2.2

Mode C — Highpass (Rf=1 kΩ, R=1 kΩ, C=100 pF, analog Fc≈1.6 MHz)

Analog signal above 25 kHz passes filter but is captured by Module 7 PCB, not MCP3208.

1.4 Assembly

3. Spray lens tube interior matte black, cure 24 hours.
4. Press lens into tube front, seal with black RTV. Mount BPW34 at focal plane (~50 mm); anode trace to TIA Pin 2 must be under 5 mm.
5. Focus: monitor TIA output on multimeter while pointing at distant light source. Slide BPW34 to peak voltage. Fix with RTV.
6. Solder decoupling caps first (100 nF across Pin 8 to GND, at pin). Solder IC, check for bridges. Solder passives, switch, photodiode last.
7. Power up before enclosing: TIA output ≈ 2.5 V with lens covered. If rail: check Cf, BPW34 polarity, Pin 2 bridge.

1.5 Software — detector.cpp

```
#include <pigpio.h>

#include <cstdio>

#include <vector>



int readADC(int h, int ch) {

   char tx[3]={(char)(0x06|(ch>>2)),(char)((ch&3)<<6),0};

   char rx[3]={0};

   spiXfer(h,tx,rx,3);

   return ((rx[1]&0x0F)<<8)|(uint8_t)rx[2];

}



int main() {
```

```
    if (gpioInitialise()<0) return 1;
    int spi=spiOpen(0,1000000,0);
    const int RATE=40000, N=16384;
    const double T_US=1e6/RATE;
    std::vector<double> s(N);
    for(int i=0;i<N;i++){
        uint32_t t0=gpioTick();
        s[i]=readADC(spi,0)*(3.3/4096.0);
        uint32_t el=gpioTick()-t0;
        if(el<(uint32_t)T_US) gpioDelay((uint32_t)T_US-el);
    }
    for(int i=0;i<N;i++) printf("%.6f\n",s[i]);
    spiClose(spi); gpioTerminate(); return 0;
}
```

Module 2 — Channel B: Near-IR Passive Detector

TEFD4300 TIA, 850 nm longpass filter, output MCP3208 CH1. Identical topology to Channel A.

2.1 Parts

Component	Qty	Source	USD ea.	Notes
TEFD4300 NIR photodiode	1	Digi-Key	$2.00	870 nm peak
OPA2354 DIP8 (unit 2)	1	Digi-Key	$2.50	
850 nm longpass filter 25 mm	1	AliExpress	$8.00	Seat at front of lens tube
25 mm biconvex lens f=50 mm	1	AliExpress	$4.00	
30 mm OD lens tube ~60 mm	1	AliExpress	$5.00	
100 kΩ 1%	1	Digi-Key	$0.10	Rf
1 pF COG ceramic	1	Digi-Key	$0.20	Cf
Subtotal			~$23	

2.2 Wiring and Calibration

TEFD4300 cathode → +5 V; anode → OPA2354 Pin 2 [trace <5 mm]

Rf=100k, Cf=1pF in parallel, Pin 1 back to Pin 2

850 nm longpass at front of lens tube, edges sealed with black electrical tape

8. Cover: CH1 ≈ 2.5 V.
9. TV remote toward tube: 38 kHz carrier aliases in ADC data, confirming NIR path live.

Module 3 — Fabry-Pérot Cavity + WGM Microsphere

3A — Fabry-Pérot Fringe Cavity

Acoustic sensing via transmitted fringe intensity modulation. Output MCP3208 CH2.

3A.1 Parts

Component	Qty	Source	USD ea.	Notes
785 nm laser diode module 5 mW	1	AliExpress	$12.00	Collimated, focus-adjustable
4.5 mm EFL collimating lens	1	AliExpress	$6.00	
R=90% partial mirror 25 mm (M1)	1	AliExpress optics	$25.00	Input coupler
R=90% partial mirror 25 mm (M2)	1	AliExpress optics	$25.00	Output coupler, PZT mount
50/50 beamsplitter cube 25 mm	1	AliExpress	$18.00	Beam injection
Kinematic mount 25 mm x2	2	AliExpress	$28.00 ea.	
200 mm optical rail + 4 carriers	1	AliExpress	$35.00	
TEFD4300 (transmitted beam PD)	1	Digi-Key	$2.00	
OPA657 (FP TIA)	1	Digi-Key	$8.50	
PZT disc 20 mm PZT-5A	1	AliExpress	$3.00	Epoxy perimeter to M2 rear
OPA445 high-voltage op-amp	1	Digi-Key	$14.50	Heatsink required
MCP4725 DAC I2C breakout	1	Adafruit / AliExpress	$5.00	
33 Ω resistor 1% 0.5 W	2	Digi-Key	$0.15	LM317 current-set (see 3A.2)
47 Ω trimmer pot	1	Digi-Key	$0.80	Series with 33 Ω fixed
LM317 TO-220	1	Digi-Key	$0.60	

1N4148 x2	2	Digi-Key	$0.10	Laser spike protection
100 µF electrolytic 25 V	1	Digi-Key	$0.25	Across laser diode
100 Ω resistor	2	Digi-Key	$0.05	OPA445 output series
19 kΩ resistor 1%	1	Digi-Key	$0.10	OPA445 Rf
1 kΩ resistor 1%	1	Digi-Key	$0.05	OPA445 Rg
±15 V dual-rail supply	1	Amazon / bench	$25.00	OPA445 rails
Dense foam 300×300 mm	1	Hardware store	$4.00	Under baseplate
Polystyrene sheet for enclosure	1	Hardware store	$5.00	Over optical rail

3A.2 Laser Driver — LM317

CORRECTION: 50 kΩ pot as sole current-set removed. A single slip to near-zero resistance destroys the laser diode. Use 33 Ω fixed + 47 Ω trimmer in series. Minimum R_set = 33 Ω, setting hard ceiling at 37.9 mA.

LM317 Output → laser anode

LM317 Adjust → laser cathode

R_set = 33Ω fixed + 47Ω trimmer in series between Adjust and Output

Trimmer max (80Ω total): I = 1.25/80 = 15.6 mA

Trimmer min (33Ω total): I = 1.25/33 = 37.9 mA ← hard ceiling

1N4148 reverse across laser diode. 100µF across laser diode, polarity correct.

10. Set trimmer to max resistance before connecting laser.
11. Power up. Measure V across R_set: I = V/R_set. Increase slowly to 35 mA.
12. Verify spot on white card at 20 cm. Do not look into beam.

3A.3 Collimation

13. Adjust collimating lens distance until beam diameter is constant 10 cm to 1 m. Spot <2 mm, circular. Fix with RTV.

3A.4 Optical Alignment

Path: laser → collimating lens → beamsplitter cube → M1 → 50 mm gap d → M2+PZT → transmitted photodiode.

14. Mount components on rail. M1 to M2: d=50 mm. Beamsplitter 30 mm before M1.
15. Align M1 normal to beam: reflected spot returns within 2 mm of laser aperture.
16. Align M2: return beam retraces through M1. Card at M1 shows coincident return spots.

17. Mount TEFD4300+OPA657 TIA (Rf=100k, Cf=1.6pF, mid-rail bias) at M2 output. Connect to oscilloscope and CH2.
18. Turn M2 adjustment screw ~1/16 turn per second. Fringes appear as oscillating transmitted intensity. If absent: re-check M2 retroreflection and iterate.

3A.5 PZT Driver — OPA445

CORRECTION: Previous version stated 0–66 V output from ±15 V rails. The OPA445 output swing on ±15 V is ±13.5 V (≈27 Vpp). Achieving 66 V output requires supply rails of at least +70 V / 0 V.

Parameter	Value	Notes
OPA445 supply	±15 V	Minimum; heatsink to TO-99 case
Gain	20 (Rf=19k, Rg=1k)	Non-inverting
Input	0–3.3 V from MCP4725	
Output swing at ±15 V	±13.5 V (≈27 Vpp)	Hard limit of supply rails
PZT drive	~27 Vpp	Sufficient for fringe sweep and lock
For higher stroke	Increase supply toward ±45 V	OPA445 rated ±150 V

OPA445 Pin 3 (non-inv.) → MCP4725 output

OPA445 Pin 2 (inv.) → output node via Rg=1k to GND

OPA445 Pin 6 (out) → 100Ω series → PZT disc

Pin 7 (+supply) → +15 V with 100 nF to GND

Pin 4 (−supply) → −15 V with 100 nF to GND

3A.6 Fringe Lock — fp_lock.cpp

```
#include <pigpio.h>

#include <cstdio>


void setDAC(int h, int v) {

    char b[2]={(char)((v>>8)&0x0F),(char)(v&0xFF)};

    i2cWriteDevice(h,b,2);

}
```

```
int readADC(int spi,int ch){
    char tx[3]={(char)(0x06|(ch>>2)),(char)((ch&3)<<6),0};
    char rx[3]={0};
    spiXfer(spi,tx,rx,3);
    return ((rx[1]&0x0F)<<8)|(uint8_t)rx[2];
}
int main(){
    gpioInitialise();
    int spi=spiOpen(0,1000000,0);
    int dac=i2cOpen(1,0x62,0);
    const int TARGET=2048;
    double Ki=0.008, dac_f=2048;
    for(;;){
        double err=readADC(spi,2)-TARGET;
        dac_f=fmax(0,fmin(4095,dac_f+Ki*err));
        setDAC(dac,(int)dac_f);
        gpioDelay(100000);  // 10 Hz control loop
    }
}
```

Note: Enclose optical rail in polystyrene box. Reduces thermal drift ~5x. Adjust Ki if lock oscillates (reduce) or cannot hold against drift (increase).

3B — WGM Microsphere Upgrade

Whispering-gallery mode resonator replaces or supplements the FP signal mirror. The microsphere is coupled via a tapered single-mode fibre. Acoustic perturbations shift WGM resonance frequencies, appearing as intensity modulation in the tapered fibre transmission when the probe laser is parked on the flank of a resonance dip. Resonant photon buildup increases effective interaction length by a factor equal to the cavity Q ($\leq 10^8$ for silica), giving orders-of-magnitude higher sensitivity than a single-pass measurement.

3B.1 Required Equipment

Component	Qty	Source	USD ea.	Notes

SM780-HP single-mode fibre, 2 m	1	Thorlabs	$22.00	Single-mode at 785 nm
Fibre stripper (250 µm coating)	1	Thorlabs T06S13	$45.00	
Micro-torch, hydrogen-oxygen or butane	1	Spirig Mini-Torch or equiv.	$35.00	Fine adjustable tip, <2 mm flame
Isopropanol 99%, lint-free wipes	1	Lab supply	$8.00	Fibre cleaning
Fibre cleaver	1	Thorlabs FK11 or equiv.	$85.00	For clean fibre end-faces
Translation stage, 3-axis, 50 nm resolution or better (x2)	2	Newport 561D-XYZ or Thorlabs NanoMax	Lab equipment	Microsphere + taper positioning
Piezo actuator for fine approach	1	Thorlabs PAZ005 or equiv.	Lab equipment	Sub-100 nm coupling gap control
Optical microscope 40x–100x	1	Lab equipment	Lab equipment	Sphere inspection and taper monitoring
Fibre launch + power meter	1	Lab equipment	Lab equipment	Monitor taper transmission during pull
FC/APC fibre adaptor x2	2	Thorlabs	$12.00 ea.	Connect fibre to laser and detector
UV-cure epoxy (Norland NOA61)	1	Thorlabs	$18.00	Fix sphere to holder if needed

3B.2 Microsphere Fabrication

Fabricate at the tip of a cleaved SM780-HP fibre. Surface tension during melting forms the sphere.

19. Strip coating from the last 3 cm of fibre using fibre stripper. Do not nick the cladding.
20. Wipe bare fibre with isopropanol-soaked lint-free wipe. Wipe in one direction only. Repeat until clean (no residue under microscope).
21. Cleave fibre at the stripped end. Inspect cleave face under microscope: must be flat and perpendicular within 1°. Recleave if chipped or angled.
22. Mount fibre vertically in a clamp with the cleaved end pointing down. Attach a small weight (0.5–1 g) to the lower end to maintain light tension during melting.
23. Apply torch tip to the cleaved fibre end for 0.3–0.6 seconds. Surface tension draws the melted glass into a sphere. Longer contact time increases sphere diameter.
24. Allow to cool undisturbed for 30 seconds. Inspect under microscope.
25. Target sphere diameter: 100–300 µm. Reject if: non-spherical, asymmetric, has satellite droplets, or has visible surface cracks.

26. Record sphere diameter from microscope reticle. WGM free spectral range: $\Delta\nu \approx c/(n\pi D)$, where n=1.45 (silica), D=sphere diameter. For D=200 µm: $\Delta\nu \approx 330$ GHz.

Note: Several attempts are normal. Flame exposure time is the primary control variable. A shorter, hotter flame (oxygen-hydrogen) gives more reproducible results than butane alone. Keep a log of exposure times vs resulting diameters.

3B.3 Tapered Fibre Fabrication

The tapered coupler transfers light evanescently into the sphere at the taper waist. Waist diameter must be comparable to the evanescent field decay length: 1–2 µm for 785 nm light in silica.

27. Strip 4 cm of coating from the centre of a 30 cm length of SM780-HP. Clean with isopropanol.
28. Mount the fibre horizontally between two translation stages separated by 25 mm. The stripped region must be centred between the stages. Apply light tension: ~0.05–0.1 N (enough to straighten the fibre without stretching it at room temperature).
29. Connect one fibre end to the 785 nm laser (via FC/APC adaptor) and the other end to the power meter. Record initial transmitted power.
30. Apply the torch flame to the centre of the stripped region. Use the narrowest, most controlled flame available. Sweep the flame back and forth over a 6–8 mm region at ~0.5 mm/s while simultaneously pulling the two translation stages apart at ~0.1 mm/s each (0.2 mm/s total elongation rate).
31. Monitor transmitted power continuously. Power will remain high during the early taper and then begin to decrease as the waist thins toward single-mode cutoff.
32. Stop pulling when transmitted power drops to 85–90% of initial. This corresponds to a waist of approximately 1–2 µm. Pulling further reduces transmission and risks fibre breakage.
33. Cool undisturbed 60 seconds. Do not move the fibre. Inspect waist region under microscope: smooth, symmetric taper with no visible kinks or nodes.
34. Test taper transmission with laser: >80% is acceptable. <70%: consider re-fabricating. Small dust contamination is the most common cause of unexplained loss; keep fibre covered when not in use.

WARNING: The taper waist is mechanically extremely fragile. Mount in a v-groove or on a dedicated fibre taper holder before any further manipulation. Do not touch the waist region with any solid object.

3B.4 Coupling Alignment

The coupling gap between the taper waist and the sphere equator must be in the range 0–500 nm for evanescent coupling. This requires piezo-driven translation.

35. Mount the microsphere fibre stub in the first 3-axis translation stage such that the sphere is accessible from the side.
36. Mount the tapered fibre in the second translation stage in a v-groove holder. Orient the taper waist parallel to and approaching the sphere equator.
37. Connect the taper input to the 785 nm laser. Connect taper output to the OPA657 TIA or a dedicated photodetector. Monitor transmission on oscilloscope.
38. Under microscope observation, manually translate the sphere toward the taper until the sphere is within ~10 µm of the taper waist (visible proximity under microscope).
39. Switch to the piezo actuator for fine approach. Advance the sphere toward the taper in 100 nm increments. At coupling onset (gap ~200–500 nm), transmission will begin to show sharp periodic dips as the laser sweeps through WGM resonances, or, with a fixed-frequency laser, transmission will drop abruptly at a specific gap distance.
40. Optimise the gap for maximum fringe contrast: transmission dips should be as deep as possible. Critical coupling occurs when the coupling rate equals the intrinsic cavity loss rate; at this point transmission reaches a minimum at resonance. This is the optimal operating point.

41. Park the laser frequency on the flank of the deepest resonance dip (half-maximum transmission). Acoustic perturbations of the sphere now appear as intensity modulation of the taper transmission. Connect taper output TIA to MCP3208 CH2 (same channel as FP output; use toggle switch to select FP or WGM path).

Note: Temperature and mechanical stability at the coupling junction are critical. Airflow, acoustic vibration, and thermal drift all modulate the gap and introduce noise. Enclose the entire coupling assembly in a draught-free box. A Peltier-stabilised baseplate improves long-term coupling stability significantly.

3B.5 Probe Laser Frequency Tuning for WGM

To find a WGM resonance with a fixed-wavelength diode laser, vary the laser frequency by changing the injection current. The tuning coefficient for a 785 nm diode is typically 0.003–0.01 nm/mA (frequency tuning ~1–5 GHz/mA). For a sphere D=200 µm with FSR=330 GHz, a 2 mA current sweep covers approximately 2–10 GHz — a small fraction of one FSR. To scan a full FSR by current tuning alone, you would need a sphere with FSR ≤10 GHz, corresponding to D ≥ 6.6 mm, which is impractical for evanescent coupling. The realistic approach is to combine current tuning (fine, ~GHz range) with temperature tuning of the laser (coarse, via TEC or heatsink temperature): temperature coefficient is ~0.3 nm/°C (~150 GHz/°C) for 785 nm diodes, so a 2–3 °C temperature shift sweeps a full FSR of a 200 µm sphere. Set the operating temperature to place a WGM resonance within the current-tuning range, then use current for fine positioning to the resonance flank.

42. With the taper at coupling distance (gap ~300 nm), slowly sweep the laser current through ±5 mA from the baseline operating point (covering ~10–25 GHz frequency range). Monitor taper transmission on oscilloscope.
43. WGM resonances appear as sharp transmission dips, width determined by cavity Q. A high-Q mode will have a dip width <0.1 mA in current sweep.
44. Select the deepest, sharpest dip. Set operating current to the half-maximum point on the low-frequency flank (decreasing current side). This is the maximum-sensitivity operating point.
45. Lock laser current to this operating point using the LM317 trimmer. Fine-lock via the 47 Ω trimmer (each degree of rotation corresponds to approximately 0.5 mA and ~100 GHz frequency shift).

Module 4 — RF Layer: RTL-SDR

100 kHz – 1.766 GHz, 2.4 MHz instantaneous bandwidth. Independent of RPi analog path.

4.1 Parts

Component	Qty	Source	USD ea.	Notes
RTL-SDR Blog V4 (incl. dipole kit)	1	rtl-sdr.com	$39.95	
FM bandstop filter (optional)	1	rtl-sdr.com	$14.95	Near broadcast towers only
LNA4ALL wideband LNA (optional)	1	AliExpress	$18.00	Bias-T powered from V4
Subtotal			~$40–55	

4.2 Driver Installation

```
sudo apt install rtl-sdr gqrx-sdr

echo 'SUBSYSTEM=="usb",ATTRS{idVendor}=="0bda",MODE="0666"' | sudo tee /etc/udev/rules.d/rtl-sdr.rules

sudo udevadm control --reload-rules && sudo udevadm trigger

rtl_test -t
```

Windows

46. Zadig as Administrator → List All Devices → Bulk-In Interface 0 → WinUSB → Replace Driver.
47. SDR# from airspy.com/downloads, run install-rtlsdr.bat.

4.3 Antenna Length

Quarter-wave monopole (cm) = 7500 / frequency (MHz)

Parameter	Value	Notes
FM broadcast	100 MHz	75 cm
VHF aircraft	130 MHz	58 cm
L-band NOAA	137 MHz	55 cm
ISM 433 MHz	433 MHz	17 cm
ADS-B	1090 MHz	6.9 cm

4.4 ELF Envelope Detector — elf_detector.cpp

```
// Build: g++ -O2 elf_detector.cpp $(gnuradio-config-info --libs) -o elf_detector

#include <gnuradio/top_block.h>

#include <gnuradio/analog/am_demod_cf.h>

#include <gnuradio/blocks/file_sink.h>

#include <osmosdr/source.h>

const double CARRIER_FREQ = 24000.0;

const double SAMP_RATE    = 250000.0;

int main(){

    auto tb=gr::make_top_block("ELF");

    auto src=osmosdr::source::make("rtl=0");

    src->set_sample_rate(SAMP_RATE);
```

```
    src->set_center_freq(CARRIER_FREQ);

    src->set_gain(30);

    auto dem=gr::analog::am_demod_cf::make(SAMP_RATE,1,20.0,100.0);

    auto snk=gr::blocks::file_sink::make(sizeof(float),"elf_out.f32");

    tb->connect(src,0,dem,0);

    tb->connect(dem,0,snk,0);

    tb->run();

}
```

Module 5 — Power Line Detection Arm

Mains current via CT probe. Isolated. Measures 50/60 Hz and harmonics to ADC Nyquist. Output MCP3208 CH3.

5.1 Parts

Component	Qty	Source	USD ea.	Notes
YHDC SCT-013-000 split-core CT	1	Amazon / AliExpress	$12.00	100A:50mA
100 Ω burden resistor 1% 0.5 W	1	Digi-Key	$0.10	Solder across CT output FIRST
OPA2354 DIP8 (unit 3)	1	Digi-Key	$2.50	Pre-amp
18 kΩ 1%	1	Digi-Key	$0.10	Rf
1 kΩ 1%	2	Digi-Key	$0.05	Rg + input series
10 µF electrolytic x2	2	Digi-Key	$0.20	DC-blocking
100 nF film cap x2	2	Digi-Key	$0.20	Rolloff + decoupling
Project box 80×50×30 mm	1	AliExpress	$3.00	
Subtotal			~$18	

WARNING: Solder the 100 Ω burden resistor across CT terminals BEFORE clamping to any live conductor.

5.2 Circuit

CT terminal 1 → 100Ω burden → CT terminal 2

Mid-rail bias: two 10k from +5V to GND; midpoint to one burden terminal; 10µF to GND

10µF DC-blocking in series on output

OPA2354 non-inv amp: Rf=18k+1k series=19k, Rg=1k, Gain=20

100nF || Rf: Fc≈83 kHz rolloff. 10µF DC-block at output. Output → CH3

5.3 CT Mounting

48. Clamp around ONE wire. Do not clamp both conductors simultaneously.
49. Verify: multimeter AC mV across burden. 1–2 A load → 50–100 mV AC.

5.4 Software — powerline_monitor.cpp

```
#include <pigpio.h>

#include <cstdio>, <vector>

int readADC(int spi,int ch){char tx[3]={(char)(0x06|(ch>>2)),(char)((ch&3)<<6),0};char
rx[3]={0};spiXfer(spi,tx,rx,3);return ((rx[1]&0x0F)<<8)|(uint8_t)rx[2];}

int main(){

    gpioInitialise(); int spi=spiOpen(0,1000000,0);

    const int RATE=5000,N=4096; const double T_US=1e6/RATE;

    std::vector<double> c(N);

    for(int i=0;i<N;i++){

        uint32_t t0=gpioTick();

        c[i]=readADC(spi,3)*(3.3/4096.0)/(20.0*100.0);  // amps

        uint32_t el=gpioTick()-t0;

        if(el<(uint32_t)T_US) gpioDelay((uint32_t)T_US-el);

    }

    for(int i=0;i<N;i++) printf("%.8f\n",c[i]);

    spiClose(spi); gpioTerminate();

}
```

Module 6 — LiFi Transmission Arm

OOK optical data link. Transmitter: Arduino C++. Receiver: RPi C++ with edge-synchronised UART decoder and per-byte bit period tracking, eliminating fixed-SPB drift over extended streams.

6.1 Parts

Component	Qty	Source	USD ea.	Notes
Arduino Nano	1	AliExpress	$6.00	
TC4427 dual MOSFET driver	1	Digi-Key	$1.50	
IRLZ44N MOSFET TO-220	1	Digi-Key	$0.80	
1W white LED	1	AliExpress	$1.00	
18–22 Ω resistor 1W	1	Digi-Key	$0.20	
10 Ω resistor	1	Digi-Key	$0.05	Gate series
100 nF ceramic	1	Digi-Key	$0.08	
20 mm collimating lens	1	AliExpress	$4.00	
Subtotal			~$14	

6.2 Circuit

Arduino D9 → 10Ω → TC4427 input

TC4427 output → IRLZ44N gate

IRLZ44N source → GND; drain → LED cathode

LED anode → 18–22Ω 1W → +5V

100 nF across +5V at drain

6.3 Transmitter — lifi_transmit.ino (Arduino C++)

```
const int  LED_PIN = 9;

const long BAUD    = 4800;

const long BIT_US  = 1000000L / BAUD;   // 208 us per bit


void sendByte(uint8_t b) {

    digitalWrite(LED_PIN, LOW);          // start bit
```

```
    delayMicroseconds(BIT_US);

    for (int i = 0; i < 8; i++) {

        digitalWrite(LED_PIN, (b >> i) & 1);

        delayMicroseconds(BIT_US);

    }

    digitalWrite(LED_PIN, HIGH);          // stop bit

    delayMicroseconds(BIT_US);

}

void setup(){ pinMode(LED_PIN,OUTPUT); digitalWrite(LED_PIN,HIGH); }

void loop() { const char* msg="POLYCHROME"; for(int i=0;msg[i];i++) sendByte(msg[i]);
delayMicroseconds(BIT_US*4); }
```

6.4 Receiver with Clock Recovery — lifi_receive.cpp

Edge-synchronised sampling with per-byte bit period estimator. At the start-bit falling edge, the sample clock resets to the measured edge time. After each successfully decoded byte, the nominal bit period is updated by blending the measured inter-byte edge timing. This eliminates accumulated clock skew between the Arduino 16 MHz crystal and the RPi system clock over long continuous streams.

```
// Build: g++ -O2 lifi_receive.cpp -lpigpio -lrt -o lifi_receive && sudo ./lifi_receive

#include <pigpio.h>

#include <cstdio>

#include <cmath>



static int readADC(int spi,int ch){

    char tx[3]={(char)(0x06|(ch>>2)),(char)((ch&3)<<6),0};

    char rx[3]={0};

    spiXfer(spi,tx,rx,3);

    return ((rx[1]&0x0F)<<8)|(uint8_t)rx[2];

}



int main(){

    if(gpioInitialise()<0) return 1;
```

```
int spi=spiOpen(0,1000000,0);


// --- Configuration ---
const int    BAUD_NOMINAL = 4800;
const int    SAMPLE_RATE  = 50000;       // 50 kSPS via pigpio
const int    THRESHOLD    = 2100;        // ADC counts; calibrate empirically
const double T_SAMPLE_US  = 1e6 / SAMPLE_RATE;


// Bit period in samples, starts at nominal, adapts per byte
double bit_period = (double)SAMPLE_RATE / BAUD_NOMINAL;  // ~10.4 samples
const double ALPHA = 0.1;  // bit period adaptation rate (0=no adapt, 1=instant)


// --- State machine ---
enum State { IDLE, CONFIRM_START, DATA, STOP };
State   state       = IDLE;
int     bit_count   = 0;
uint8_t byte_acc    = 0;
double  next_sample = 0;  // sample index of next mid-bit sample
double  sample_idx  = 0;  // running sample counter (fractional)
uint32_t start_edge_tick = 0;  // gpioTick() at start-bit falling edge
int     prev_raw    = 1;  // previous bit level (mark=1)


for(;;){
    // --- Timed ADC read ---
    uint32_t t0=gpioTick();
    int raw=readADC(spi,0);
    int bit=(raw>THRESHOLD)?1:0;


    switch(state){
    case IDLE:
```

```
    // Detect falling edge (mark->space = start-bit onset)
    if(prev_raw==1 && bit==0){
        start_edge_tick = t0;
        // Next sample point: 1.5 bit-periods from edge (mid of bit 0)
        next_sample = sample_idx + bit_period * 1.5;
        state = CONFIRM_START;
    }
    break;

case CONFIRM_START:
    if(sample_idx >= next_sample){
        if(bit==0){  // still space: valid start bit
            bit_count = 0;
            byte_acc  = 0;
            next_sample += bit_period;  // advance to bit 0 mid-point
            state = DATA;
        } else {
            state = IDLE;  // false edge, abort
        }
    }
    break;

case DATA:
    if(sample_idx >= next_sample){
        byte_acc |= (uint8_t)(bit << bit_count);
        next_sample += bit_period;
        if(++bit_count == 8) state = STOP;
    }
    break;
```

```
        case STOP:
            if(sample_idx >= next_sample){
                if(bit==1){  // valid stop bit
                    printf("%c", (char)byte_acc);
                    fflush(stdout);
                    // --- Clock recovery: measure actual bit period from this byte ---
                    // Total byte duration = 10 bit-times (1 start + 8 data + 1 stop)
                    uint32_t elapsed_us = gpioTick() - start_edge_tick;
                    double measured_bp  = (double)elapsed_us / (10.0 * T_SAMPLE_US);
                    // Sanity check: reject if >20% off from current estimate
                    if(fabs(measured_bp - bit_period) / bit_period < 0.20){
                        bit_period = (1.0-ALPHA)*bit_period + ALPHA*measured_bp;
                    }
                }
                state = IDLE;
            }
            break;
        }


        prev_raw = bit;
        sample_idx += 1.0;


        // --- Maintain sample rate ---
        uint32_t el=gpioTick()-t0;
        if(el<(uint32_t)T_SAMPLE_US) gpioDelay((uint32_t)T_SAMPLE_US-el);
    }
}
```

Note: SPB at 50 kSPS / 4800 baud = 10.4. The ALPHA=0.1 adaptation rate allows the estimated bit period to track slow crystal drift without over-reacting to noise. Calibrate THRESHOLD: run with LED covered (record noise floor level), then uncover (record signal level). Set THRESHOLD to the midpoint. Increase BAUD to 9600 only after confirming stable decode at 4800.

Module 7 — MHz Optical Detection: OPA657 TIA PCB + Red Pitaya

Full 1–100 MHz optical detection path. Two-layer component: (1) OPA657 TIA on a dedicated 4-layer PCB with 50 Ω output, and (2) Red Pitaya STEMlab 125-14 as 125 MSPS digitiser. Channel A Mode C analog front-end connects to this module via SMA cable.

7.1 Parts

Component	Qty	Source	USD ea.	Notes
Red Pitaya STEMlab 125-14	1	redpitaya.com	$349.00	125 MSPS, 14-bit, ±1 V input, FPGA, Ethernet
OPA657 SOT-23-5 (U1)	1	Digi-Key	$8.50	GBW 1.6 GHz TIA; SOT-23-5 pkg for RF layout
BPW34S (SMD photodiode)	1	Digi-Key	$1.40	SMD version of BPW34; 550 nm peak
100 Ω SMD 0402 resistor	2	Digi-Key	$0.05 ea.	Output 50Ω series termination
1 kΩ SMD 0402 resistor (Rf)	1	Digi-Key	$0.05	TIA feedback
10 kΩ SMD 0402 (x2, mid-rail)	2	Digi-Key	$0.05	Bias divider
0.5 pF COG 0402 (Cf)	1	Digi-Key	$0.20	TIA stability
100 nF 0402 ceramic (x4)	4	Digi-Key	$0.08 ea.	Supply decoupling
10 nF 0402 ceramic	2	Digi-Key	$0.08 ea.	Additional HF decoupling
1 nF 0402 ceramic	2	Digi-Key	$0.08 ea.	HF decoupling
SMA edge-launch connector (x2)	2	Digi-Key (Amphenol 132289)	$3.50 ea.	Input (PD) and output (to Red Pitaya)
Ferrite bead 0402 600Ω@100 MHz	2	Digi-Key	$0.15	Supply isolation on board
PCB fabrication (4-layer, 30×45 mm)	1	JLCPCB / OSH Park	~$15–25	See layout spec below
SMA cable male-male 30 cm	1	AliExpress	$6.00	TIA output to Red Pitaya IN1
Subtotal			~$415	

7.2 PCB Layout Specification

This is an RF-grade design. Standard perfboard will not work above ~5 MHz due to parasitic inductance and capacitance. The PCB must be ordered and assembled before this module is functional.

Stackup

Parameter	Value	Notes
Layer 1 (top)	Signal + components	35 µm copper
Prepreg	FR4, 0.2 mm	Standard 4-layer prepreg
Layer 2	Continuous ground plane, no cuts	35 µm copper
Core	FR4, 1.0 mm	
Layer 3	Power plane (+5 V pour, GND pour)	35 µm copper
Prepreg	FR4, 0.2 mm	
Layer 4 (bottom)	Ground fill, minor routing	35 µm copper

Note: Order 4-layer stackup from JLCPCB (JLC04161H-3313 or equivalent) or OSH Park 4-layer. Total board thickness 1.6 mm. Specify impedance control: 50 Ω microstrip on Layer 1 over Layer 2 ground. At FR4 εr=4.4, 0.2 mm dielectric: 50 Ω trace width = 0.38 mm. Confirm with fab impedance calculator before submitting Gerbers.

Component Placement Rules

- OPA657 (U1): centre of board. Orientation: output pin toward output SMA, inverting input toward input SMA. No right-angle trace bends anywhere on board.
- Rf (1 kΩ) and Cf (0.5 pF) in parallel: place within 0.5 mm of OPA657 inverting input and output pins. These are the highest-priority placements on the board.
- Input SMA: edge-launch, connected directly to BPW34S cathode (anode to +5 V via 10 kΩ). Alternatively, BPW34S mounts on the PCB adjacent to the input SMA with a short stub trace. Photodiode anode-to-TIA trace must be under 3 mm.
- Output SMA: edge-launch, connected to OPA657 output via 50 Ω microstrip then 100 Ω series resistor (two 100 Ω in parallel = 50 Ω termination matching).
- Decoupling hierarchy at each supply pin: 1 nF closest, then 10 nF, then 100 nF, spaced radially outward. Vias to ground plane as short as possible (no via stubs).
- Guard ring: continuous copper ring on Layer 1 connected to Layer 2 ground, surrounding the inverting-input node trace and Rf/Cf components. Single via to ground plane at one point of the ring.
- Mid-rail bias: two 10 kΩ from +5 V to GND, midpoint decoupled with 100 nF to GND, connected to OPA657 non-inverting input. Place mid-rail RC on Layer 1, outside the guard ring.

Trace Routing

- All signal traces on Layer 1: no signal routing on Layer 2 (ground plane must be unbroken).

- Input trace from SMA to OPA657 Pin 2 (inv.): <5 mm, 50 Ω microstrip (0.38 mm width over 0.2 mm prepreg). Surrounded by ground on both sides.
- Output trace from OPA657 Pin 6 to output SMA: 50 Ω microstrip. Route straight, no bends sharper than 45°.
- Power traces: 0.5 mm width minimum. Short paths to decoupling caps.
- No trace crosses over a via or gap in the ground plane.

KiCad Design Notes

50. Create 4-layer board (File → Board Setup → Layers). Assign: F.Cu = signal, In1.Cu = GND, In2.Cu = PWR, B.Cu = GND fill.
51. Set board outline 30 mm × 45 mm.
52. Add copper fill on In1.Cu (ground plane): pour, no thermal reliefs, connect to GND net.
53. Set design rules: min trace 0.2 mm, min clearance 0.2 mm, min via drill 0.3 mm.
54. Run DRC before Gerber export. Zero errors and zero unconnected nets required.
55. Export Gerbers: Gerber → Plot (F.Cu, B.Cu, In1.Cu, In2.Cu, F.Mask, B.Mask, F.Silkscreen, Edge.Cuts). Export drill file (Excellon format). ZIP all files for JLCPCB upload.

7.3 Assembly

56. Solder OPA657 first using fine-tip iron at 320°C or reflow paste + hotplate. Inspect under microscope for bridges.
57. Solder Rf and Cf immediately after OPA657, within same session to avoid handling stress on fine-pitch pads.
58. Solder all 0402 passives. Decoupling caps in order: 1 nF, 10 nF, 100 nF at each supply node.
59. Solder SMA connectors last. Apply flux before soldering, clean with isopropanol after.
60. Power-up check before connecting to Red Pitaya: +5 V on supply, mid-rail at 2.50 V at OPA657 non-inverting input, output ≈ 2.50 V with no light input (DC-biased). If output is stuck at rail: check Rf/Cf, verify non-inverting pin voltage.

7.4 Red Pitaya Setup

61. Connect Red Pitaya to network via Ethernet. Default IP: 192.168.128.1 (direct connection) or DHCP (via router). Confirm ping response.
62. SSH: ssh root@<ip> (default password: root). Change password immediately.
63. Connect TIA output SMA to Red Pitaya IN1 (50 Ω input, ±1 V full-scale, AC or DC coupled selectable in software).
64. Set input coupling to DC. The OPA657 TIA output is centred at mid-rail (2.5 V) with AC signal on top. Use a DC-blocking capacitor (10 nF SMD, soldered inline on the SMA cable at the Red Pitaya end) if the 2.5 V DC offset saturates the input. Red Pitaya IN1 DC range is ±1 V; mid-rail is 2.5 V, so DC blocking is required.

Note: Add a 10 nF 0402 ceramic in series on the SMA cable at the Red Pitaya IN1 connector. This blocks the 2.5 V mid-rail DC offset and passes signals from ~320 kHz upward ($Fc = 1/(2\pi \times 50\ \Omega \times 10\ nF) = 318$ kHz). For signals below 318 kHz, increase the blocking cap to 100 nF (Fc = 31.8 kHz).

7.5 Acquisition Software — rp_acquire.cpp

Connects to Red Pitaya via SCPI over TCP. Commands the ADC, retrieves waveform data, writes to file. Run on any Linux host with network access to the Red Pitaya. Requires no special libraries beyond POSIX sockets.

// Build: g++ -O2 rp_acquire.cpp -o rp_acquire

```
// Run:   ./rp_acquire 192.168.1.100 capture.bin


#include <sys/socket.h>

#include <arpa/inet.h>

#include <netinet/in.h>

#include <unistd.h>

#include <cstdio>

#include <cstring>

#include <cstdlib>

#include <vector>


static int g_sock = -1;


static void scpi_send(const char* cmd){

    char buf[256];

    snprintf(buf, sizeof(buf), "%s\r\n", cmd);

    send(g_sock, buf, strlen(buf), 0);

}


static std::string scpi_query(const char* cmd){

    scpi_send(cmd);

    char buf[65536]; int n=0;

    while((n=recv(g_sock, buf, sizeof(buf)-1, 0))>0){

        buf[n]=0;

        if(strchr(buf,'\n')) break;

    }

    return std::string(buf);

}


int main(int argc, char** argv){
```

```
if(argc < 3){ fprintf(stderr,"Usage: rp_acquire <ip> <outfile>\n"); return 1; }
const char* rp_ip   = argv[1];
const char* outfile = argv[2];

// Connect to Red Pitaya SCPI port 5000
g_sock = socket(AF_INET, SOCK_STREAM, 0);
struct sockaddr_in sa{};
sa.sin_family      = AF_INET;
sa.sin_port        = htons(5000);
inet_pton(AF_INET, rp_ip, &sa.sin_addr);
if(connect(g_sock,(struct sockaddr*)&sa,sizeof(sa))<0){
    perror("connect"); return 1;
}

// --- Configure acquisition ---
scpi_send("ACQ:RST");
scpi_send("ACQ:SRAT 125M");          // 125 MSPS
scpi_send("ACQ:DEC 1");              // no decimation
scpi_send("ACQ:TRIG:LEV 0.01");      // 10 mV trigger level
scpi_send("ACQ:TRIG:DLY 0");
scpi_send("ACQ:START");
scpi_send("ACQ:TRIG CH1_PE");        // trigger on IN1 rising edge

// Wait for trigger
char state[64]="";
for(int i=0;i<200;i++){
    std::string r=scpi_query("ACQ:TRIG:STAT?");
    if(r.find("TD")!=std::string::npos) break;
    usleep(10000);  // 10 ms poll
}
```

```
    // Read waveform from IN1 (16384 samples at 125 MSPS = 131 us window)
    std::string raw = scpi_query("ACQ:SOUR1:DATA?");

    // Parse CSV float values
    std::vector<float> data;
    char* p = (char*)raw.c_str();
    // Skip '#<n><nbytes>' header if present
    if(p[0]=='#'){ int nd=p[1]-'0'; p+=2+nd; }
    while(*p){ float v=strtof(p,&p); data.push_back(v); if(*p==',')p++; }

    // Write binary float32 to file
    FILE* f=fopen(outfile,"wb");
    fwrite(data.data(), sizeof(float), data.size(), f);
    fclose(f);

    printf("Captured %zu samples at 125 MSPS -> %s\n", data.size(), outfile);
    close(g_sock);
    return 0;
}
```

Note: Red Pitaya SCPI server must be running: on the Red Pitaya, run /opt/redpitaya/bin/scpi-server & or enable it via the web interface (Applications → SCPI). The 16384-sample buffer at 125 MSPS gives a 131 µs capture window. For longer captures, use decimation (ACQ:DEC 2/4/8/...) or loop acquisition. Output binary float32 file can be analysed in any FFT tool.

7.6 Expected Performance

Parameter	Value	Notes
ADC	Red Pitaya STEMlab 125-14	125 MSPS, 14-bit
Nyquist limit	62.5 MHz	Captures 1–62.5 MHz optical signals

TIA bandwidth	~10 MHz at Rf=1kΩ	GBW/Rf, OPA657 GBW=1.6 GHz
For full 62.5 MHz capture	Reduce Rf to 25 Ω	BW = 1.6 GHz/25 = 64 MHz; lower gain
Input-referred noise	~5 nV/√Hz at Rf=1kΩ	OPA657 voltage noise 4.8 nV/√Hz
Dynamic range	~72 dB	14-bit ADC
PCB required	Yes, 4-layer, impedance controlled	Perfboard not suitable above ~5 MHz

Consolidated Parts and Pricing

System Total

Module	Approx. Cost (USD)
Shared Platform (RPi, MCP3208, supply)	~$55
Module 1 — Channel A	~$32
Module 2 — Channel B passive	~$23
Module 3A — Fabry-Pérot cavity	~$195
Module 3B — WGM microsphere (consumables only)	~$75
Module 3B — WGM (lab equipment: stages, microscope)	Lab equipment
Module 4 — RTL-SDR	~$55
Module 5 — Power line CT	~$18
Module 6 — LiFi	~$14
Module 7 — MHz PCB + Red Pitaya	~$415
TOTAL (consumables + commodity hardware)	~$882
Without Module 7 (kHz-only system)	~$467
Without Modules 3B and 7 (core bench system)	~$392

Note: Module 3B WGM consumables (~$75) assumes SM780-HP fibre, torch, cleaver, and FC/APC adaptors. Translation stages, piezo controllers, and microscope are assumed to be available as lab equipment and are not costed here. Module 7 Red Pitaya ($349) dominates; this is the correct instrument for this application.

Appendix — Channel Allocation Summary

Parameter	Value	Notes

MCP3208 CH0	Module 1 Ch. A output	All modes via rotary switch
MCP3208 CH1	Module 2 Ch. B passive NIR	
MCP3208 CH2	Module 3 FP or WGM output	Toggle switch selects path
MCP3208 CH3	Module 5 power line CT	
MCP3208 CH4–7	Reserved	
Red Pitaya IN1	Module 7 OPA657 TIA output	125 MSPS, DC-blocked at connector
RTL-SDR	Module 4 RF	Independent, not on RPi SPI bus